

Assignment-2

Due on 24/09/2026

Note-1: Please submit the Programming and Non-programming (*handwritten and scanned*) component as one compressed file only.

Note-2: Students may be called for a viva voce to assess their understanding of the concepts, methodology, implementation, and results presented in the submitted assignment.

Non-programming:

[1] While obtaining the Bellman equation for the value function, it was shown that

$$v_{\pi}(s) = E_{\pi}[R_{t+1} + \gamma v_{\pi}(s') | S_t = s] \quad (1)$$

$$= \sum_{a \in A} \pi(a|s) \sum_{s' \in S} p(s'|s, a) [r(s, a, s') + \gamma v_{\pi}(s')] \quad (2)$$

where $p(s'|s, a) = P(S_{t+1} = s' | S_t = s, A_t = a)$ and $r(s, a, s') = E[R_{t+1} | S_t = s, A_t = a, S_{t+1} = s']$. Obtain all the intermediate steps between eq (1) and eq (2) using ideas of conditional expectation etc.

[2] Solve the following problems from the textbook "Sutton, R. S., & Barto, A. G. (2018). Reinforcement learning: An introduction (2nd ed.). MIT Press." link: <http://incompleteideas.net/book/the-book-2nd.html>

- Exercise 3.17
- Exercise 3.23
- Exercise 3.25
- Exercise 3.26
- Exercise 4.1
- Exercise 4.2
- Exercise 4.3
- Exercise 4.4

Programming:

[4] (Iterative Policy evaluation) Consider a 5×5 gridworld problem given in Figure 1, where the objective is to reach the terminal states denoted by black squares from any other non-terminal states. There are four actions possible in each state, $A = \{up, down, right, left\}$, which deterministically cause the corresponding state transitions, except that actions that would take the agent off the grid in fact leave the state unchanged. Consider the discount factor γ as 0.95. The reward is -1 on all transitions until the terminal state is reached. The expected reward function is thus $r(s, a, s') = -1$ for all states s, s' and action a .

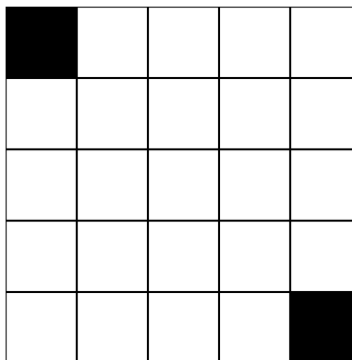


Figure 1: 5×5 gridworld problem

Suppose the agent follows the equiprobable random policy (all actions equally likely). Let the value be defined as the expected number of steps from the given state until termination. Start from the initial estimates of values as 0 in each grid and obtain the final estimates of values in each grid using Iterative Policy evaluation

$$v_{k+1}(s) = \sum_{a \in A} \pi(a|s) \sum_{s' \in S} p(s'|s, a) [r(s, a, s') + \gamma v_k(s')]$$

[5] (Value Iteration) Consider the 5×5 gridworld problem given in Figure 1, and obtain the optimal values. Also show the optimal policy graphically. Use `Grid_Bare_template.py` and `Grid_show_actions.py`, which are provided, for the purpose of displaying the actions. Both files are located at <https://drive.google.com/drive/folders/1kl9VpfbyaDfkRYTbIdcHHpN9w4jlE1Bj?usp=sharing>

[6] (Policy Iteration) Consider the 5×5 gridworld problem given in Figure 1, and obtain the optimal values. Also show the optimal policy graphically. Consider the values as 0 at the beginning of each policy evaluation step. Further, compare policy iteration with the value iteration of Problem [5]: report the number of iterations (sweeps) and the computation time taken by each method to converge, and comment on the difference.

[7] It is almost a similar environment as Figure 1. Consider a gridworld in Figure 2 with ‘terminal state’ (green, desired parking spot) with reward +10 and a ‘bad state’ (red, pothole) with reward -50, and reaching any other state lead to 0 reward. The black grid can be seen as an obstacle to avoid in the grid. Taking an action that would bump into a wall/obstacle leaves agent where it is.

0	0	0	10
0		0	-50
0	0	0	0
-50		0	0

Figure 2

The actions move in the desired direction only with probability 0.8, while the vehicle can go in two directions perpendicular to the desired direction with equal probability 0.1 each, for example the action move for ‘up’ and ‘right’ are shown in Figure 3.

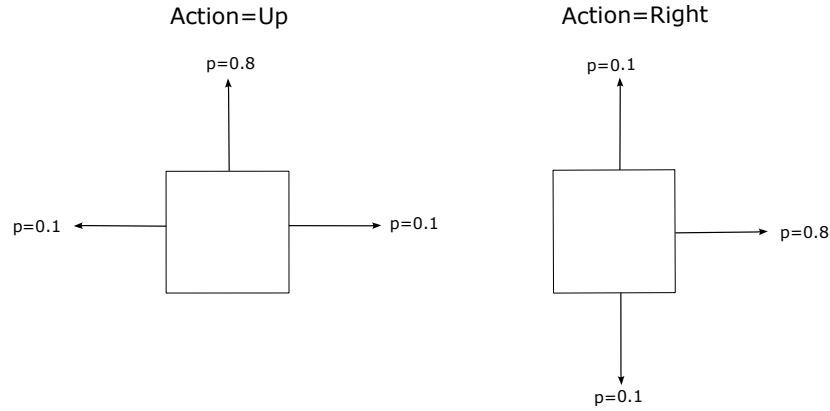


Figure 3: Schematic of the perpendicular action outcomes.

Then

- i) Using value iteration update

$$v_{k+1}(s) = \max_a \sum_{s'} p(s'|s, a) [r(s, a, s') + \gamma v_k(s')]$$

and $\gamma = 0.9$, with initial values as shown in Figure 2, obtain the values after 10 iterations.

- ii) Show the resulting policy at the end of 10th iteration
- iii) Using policy iteration equations

$$\text{policy evaluation: } v_{\pi_k}(s) = \sum_{s'} p(s'|s, a) [r(s, a, s') + \gamma v_{\pi_k}(s')]$$

$$\text{policy improvement: } \pi_{k+1}(s) = \arg \max_a \sum_{s'} p(s'|s, a) [r(s, a, s') + \gamma v_{\pi_k}(s')]$$

and the initial policy $\pi_0 = \text{Up}$ for all states, Solve V_{π_0} , π_1 , V_{π_1} , π_2 and show these on the gridworld. Again similar to the above problem, consider the values as 0 at the beginning of each policy evaluation step.